# CAREER: A Unified Framework for Differential Privacy in Multimodal Medical AI: Modeling, Federation, and Auditing

## Project Summary


### Overview

Machine learning is transforming medicine, with AI systems now deployed for clinical risk prediction, medical imaging interpretation, clinical documentation, and genomic analysis. This progress is fundamentally data-driven, yet the sensitive nature of patient data creates a profound tension between the data requirements of modern AI and the privacy rights of patients. Differential privacy (DP) has emerged as the gold standard for privacy protection, with DP-SGD becoming the canonical algorithm for training neural networks under formal privacy guarantees. However, DP remains strikingly absent from deployed medical AI. This gap reflects fundamental mismatches between existing DP algorithms and the structure of medical data, which spans heterogeneous modalities with rich correlations within and across modalities, longitudinal accumulation per patient, and deployment across multiple institutional trust boundaries.

This project develops a unified framework for differential privacy in multimodal medical AI, organized into three integrated thrusts: (1) Design a flexible privacy modeling language that exposes the secret, external constraint, neighboring relation, and privacy loss measure as first-class design parameters, paired with libraries of new and existing mechanisms. (2) Extend the framework to federated and multi-institutional settings with hierarchical trust models and site-heterogeneous federated mechanisms. (3) Develop auditing methodologies calibrated to the PML framework.

### Intellectual Merit

The project makes foundational theoretical contributions by unifying the currently fragmented landscape of DP variants into a single parameterized framework with proven compilation semantics, and by extending this framework to the correlated, multimodal, and hierarchical data structures characteristic of medicine. It develops tight privacy-utility analyses for settings where existing theory is loose or absent. It advances the theory of empirical privacy auditing beyond its current scope. The work sits at the intersection of differential privacy, machine learning, federated systems, and biomedical informatics, and will produce contributions to top venues in each of these fields. By bridging theoretical rigor with practical deployability, the project addresses open problems identified as priorities by NSF, NIH, NIST, and the broader privacy research community, while creating a coherent intellectual foundation that subsequent work in private medical AI can build upon.

### Broader Impact

By providing rigorous, flexible, and auditable privacy tools calibrated to the actual structure of medical data, this research program aims to move formal privacy from a theoretical aspiration to a deployable standard in medical AI, improving the privacy protections available to patients while enabling the clinical benefits of data-driven medicine. Open-source software released through the project, integrated with widely-used ML and federated learning frameworks, will make these tools accessible to academic medical centers, community hospitals, public health agencies, and regulatory bodies. Reference deployments developed in partnership with academic medical centers will demonstrate the framework on realistic multi-institutional medical AI tasks, providing templates that other institutions can adapt.